01-11-2026 Decimals

C&L Math Tutoring

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Decimal place values

Since we've already covered fractions, we can now explore decimals, which are another way to represent parts of a whole.

$$3,567,432.245895$$

Numbers after the decimal point represent parts of a whole, just like fractions.

Let's look at place value:

Three **millions** 10^6
Five **hundred thousands** 10^5
Six **ten thousands** 10^4
Seven **thousands** 10^3
Four **hundreds** 10^2
Three **tens** 10^1
Two **ones** 10^0
Decimal point goes here
Two **tenths** 10^{-1}
Four **hundredths** 10^{-2}
Five **thousandths** 10^{-3}
Eight **ten thousandths** 10^{-4}
Nine **hundred thousandths** 10^{-5}
Five **millionths**10^{-6}

Word form

To get the word form of a decimal:

1. Say the whole number part normally.
2. Say "and" for the decimal point.
3. For the decimal part, say it as though it were a whole number, then say the place value of the last digit.

Three million, five hundred sixty-seven thousand, four hundred thirty-two and two hundred forty-five thousand, eight hundred ninety-five millionths.

Conversions

Four *hundredths* $= \frac{4}{100} = \frac{1}{25} = 0.04$

To convert from a fraction to a decimal, perform long division of the numerator by the denominator.

To convert from a decimal to a fraction, write the decimal as a fraction with a denominator that is a power of ten (last place value), then simplify.

Adding and subtracting decimals

When adding and subtracting decimals, line up the decimal point and add/subtract as usual.

Example 1: 0.75 + 2.4

$$\begin{aligned} & 0.75 \\ + & \, 2.40 \\ = & \, 3.15 \end{aligned}$$

Example 2: 5.2 - 3.75

$$\begin{aligned} & 5.20 \\ - & \, 3.75 \\ = & \, 1.45 \end{aligned}$$

Multiplying decimals

When multiplying decimals, multiply as usual, then place the decimal point in the product so that it has the same number of decimal places as the sum of the decimal places in the factors.

$$\begin{aligned} & 0.25 \\ \times & \, 0.67 \\ & \, 175 \\ + & \, 1500 \\ = & \, 0.1675 \end{aligned}$$